Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI engineer and technical educator specializing in **LangChain, LangGraph, and agentic AI** - building **enterprise enablement** artifacts and production systems. Built **Leasel** - an 8-agent LangGraph platform for document intelligence, support investigation, and tenant self-service: RAG retrieval, confidence scoring, HITL escalation, and a reproducible eval harness (93.6% precision, 91% Retrieval Precision@1). Delivered **live technical workshops to 600+ students** at George Mason University - customer onboarding-style instruction across Python programming, AI fundamentals, and database design - weekly lab instruction and one-on-one debugging support. At Entrata (PropTech, 100K+ MAU), shipped production modules across 20+ full-stack systems on AWS across 120+ Agile sprints. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

LangChain / LangGraph / LangSmith | Agentic AI Systems | RAG Pipelines | LLM Evaluation & Observability | Technical Workshop Delivery | Customer Enablement | Reference Implementations | Enterprise Onboarding | HITL & Guardrails | Forward Deployed AI Engineering

WORK EXPERIENCE

George Mason University - College of Engineering

Aug 2025 - May 2026

Graduate Teaching Assistant (Part-time)

- Designed and delivered **live technical workshops to 600+ students** across multiple MIS 303 sections - Python programming, AI fundamentals, database design, and SQL; weekly lab instruction and office hours.
- **Python Programming:** Taught fundamentals, data structures, OOP, file I/O, and automation scripting; developed supplementary exercises and code walkthroughs; provided targeted one-on-one debugging and logic-building support.
- **Database Design:** Guided students through ER modeling, normalization, relational tables, SQL query writing, and data validation; graded assignments with rubric-aligned feedback across all sections.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an **agentic mission-planning system** as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) - RAG layer over FAA Part 107 corpus; every compliance check cited to source regulation; confidence-scored output with ChromaDB upgrade path.
- Engineered **validation guardrails** enforcing altitude limits, battery budgets, and geometric no-fly-zone checks - blocking non-compliant missions before export; improved agent accuracy through iterative prompt engineering.

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate

- Achieved $R^2 = 0.977$ across 190+ countries building multi-source prediction models (Python, R) on WHO and UK Road Safety datasets; applied Isolation Forest to surface anomalies manual review missed across 3 research pipelines.
- Improved research comprehension 40% across two cohorts by designing faculty leadership dashboards in Tableau, ggplot2, and Quarto.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer (Remote)

- **Shipped production code across 20+ full-stack modules** (PHP, Python, React, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ MAU - 99.5% uptime over 6+ consecutive months.
- **Shipped first production AI pipeline** on 1.6M real financial records (Python, PyTorch, OpenAI API) - automated data processing replacing manual Excel workflows; output validation gates, quality controls, and compliance guardrails at launch.
- **Optimized 200+ SQL queries** (30% performance gain); led microservices redesign using Docker - 50% scalability improvement, 30% deployment time reduction; AWS S3 migration with 100% PHI compliance, zero security incidents.
- Eliminated 80% of manual operational workload via Python/Bash automation pipelines; mentored 5+ junior engineers across 300+ peer code reviews.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Built full-stack property management modules (PHP, Python, PostgreSQL); ETL pipelines for 1.6M HUD/Tax Credit records; cloud migration to AWS S3; redesigned document storage database architecture.

PROJECTS

Leasel: Agentic AI Platform - Document Intelligence & Support Investigation 2026

Built an 8-agent **LangGraph** platform demonstrating production agent engineering best practices: (1) **Document Intelligence** - OCR-fallback ingestion, section-aware chunking, TF-IDF RAG retrieval, 100% citation correctness, human-review gate on regulated decisions; (2) **Support Investigation** - stateful pipeline (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → HITL Escalation) with conditional routing, confidence scoring, and evidence-backed root-cause hypotheses; (3) **Tenant Self-Service** - hard safety boundaries enforced in code. Reproducible eval harness: 93.6% precision, 91% Retrieval Precision@1, 81% pass@1, LLM-as-judge scoring. Core rule: LLM reasons; deterministic logic or human makes every final regulated call.

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · LangSmith-compatible eval harness

AI on FHIR: NLP-Driven Medical Records Query System 2024

Built a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings - reduced manual clinical retrieval time ~60% and enforced HIPAA-compliant RBAC across structured medical records.

Python · Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker

EDUCATION

M.S. Data Analytics Engineering · George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering · Jawaharlal Nehru Technological University — 2017

SKILLS

Languages: Python, PHP, JavaScript, TypeScript, SQL, PL/SQL, R, Bash

AI & Agents: LangChain, LangGraph, LangSmith, Agentic AI, Multi-Agent Orchestration, Tool Calling, RAG Pipelines, Semantic Search, Embeddings, LLM Evaluation, LLM-as-Judge, Confidence Scoring, HITL, Guardrails, Prompt Engineering, Output Validation, Hallucination Detection, OpenAI API, Anthropic API, FastAPI, ChromaDB, spaCy, BERT

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), Microservices, PostgreSQL, MySQL, Oracle, RabbitMQ, Git, Linux, Agile/Scrum

ML & Data: PyTorch, Scikit-learn, XGBoost, Random Forest, Isolation Forest, Feature Engineering, NumPy, Pandas, Tableau, Power BI, ggplot2